Assignment-Regression Algorithm

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

1.) Identify your problem statement
Using regression to find insurance

2) Tell basic info about the dataset

Input – Age, BMI, Children, Sex, Smoker, Charges

3) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

Use get_dummies for converting string to number (sex, smoker)

4) Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model

Random Forest(Criterion = "squared_error", n_estimators = 100,max_features="sqrt"), R_Score = 0.87

5) All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

Multiple Linear Regression – 0.78

SUPPORT VECTOR MACHINE:

S.NO	HYPER PARAMETER	Linear (r value)	Rbf (r value)	Poly (r value)	Sigmoid (r value)
1	C10	0.4	-0.03	0.03	0.03
2	C100	0.62	0.32	0.61	0.52
3	C500	0.76	0.66	0.82	0.44
4	C1000	0.76	0.81	0.85	0.28
5	C2000	0.74	0.85	0.86	-0.59
6	C3000	0.74	0.86	0.85	-2.12

The SVM Regression use R_Score Value (Rbf, Poly and Hyper Parameter(C3000)) = 0.86

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DECISION TREE:

SL.NO	CRITERION	MAX FEATURES	SPLITTER	R VALUE
1	Squared Error	None	Best	0.68
2	Squared Error	None	Random	0.70
3	Squared Error	Sqrt	Best	0.35
4	Squared Error	Sqrt	Random	0.73
5	Squared Error	Log2	Best	0.67
6	Squared Error	Log2	Random	0.64
7	Absolute Error	None	Best	0.68
8	Absolute Error	None	Random	0.75
9	Absolute Error	Sqrt	Best	0.71
10	Absolute Error	Sqrt	Random	0.67
11	Absolute Error	Log2	Best	0.44
12	Absolute Error	Log2	Random	0.66
13	Poisson	None	Best	0.72
14	Poisson	None	Random	0.74
15	Poisson	Sqrt	Best	0.75
16	Poisson	Sqrt	Random	0.72
17	Poisson	Log2	Best	0.66
18	Poisson	Log2	Random	0.69

The Decision Tree Regression use R_Score Value (Absolute Error, None and Random) = 0.75

6.) Mention your final model, justify why u have chosen the same.
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RANDOM FOREST:

SL.NO	CRITERION	MAX FEATURES	N_ESTIMATORS	R VALUE
1	Squared Error	None	10	0.83
2	Squared Error	None	100	0.85
3	Squared Error	Sqrt	10	0.85
4	Squared Error	Sqrt	100	0.87
5	Squared Error	Log2	10	0.85
6	Squared Error	Log2	100	0.87
7	Absolute Error	None	10	0.83
8	Absolute Error	None	100	0.84
9	Absolute Error	Sqrt	10	0.85
10	Absolute Error	Sqrt	100	0.86
11	Absolute Error	Log2	10	0.85
12	Absolute Error	Log2	100	0.86
13	Poisson	None	10	0.83
14	Poisson	None	100	0.85
15	Poisson	Sqrt	10	0.85
16	Poisson	Sqrt	100	0.86
17	Poisson	Log2	10	0.85
18	Poisson	Log2	100	0.86

The Decision Tree Regression use R_Score Value (Squared Error, sqrt and 100) = 0.87

6.) Mention your final model, justify why u have chosen the same.
Random Forest